

2023-Ph-H1-Q18

Section: Electricity

Topic: Current, PD, Power, Resistance

Which circuit has the smallest total resistance?

Answer: C

(Two $4\ \Omega$ resistors in parallel)

Guidance for Students:

Parallel resistors lower the total resistance.

In parallel:

$$\frac{1}{R_{\text{total}}} = \frac{1}{4} + \frac{1}{4} = \frac{1}{2} \Rightarrow R_{\text{total}} = 2\ \Omega$$

Revision Tips:

- Series: resistances add.
- Parallel: total resistance is **less than smallest resistor**.
- Use parallel formula!